

CSA1280 – COMPUTER ARCHITECTURE

EXPERIMENT – 33

FULL ADDER USING OR GATE

AIM: To design and implement a Full Adder using OR gate circuit using Logisim simulator.

ALGORITHM

1. Open the Logisim simulator.
2. Place three input pins: A, B, and Cin.
3. Use XOR gates to generate the Sum.
4. Use AND gates to generate the partial carry terms.
5. Use OR gates to combine the carry terms and obtain Carry-out.
6. Connect the outputs to Sum (S) and Carry-out (Cout).
7. Test all possible input combinations.
8. Verify the outputs using the truth table.

Truth Table:

A	B	Cin	Sum (S)	Carry-out
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

RESULT

